13 (a) Write $\frac{5}{x+1} + \frac{2}{3x}$ as a single fraction in its simplest form.

Common denominator = 3x(x + 1).

5/(x + 1) + 2/(3x)

= [15x + 2(x + 1)] / [3x(x + 1)]

= (17x + 2) / [3x(x + 1)].

2 marks total
Use 3x(x + 1)
Simplify numerator

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(2)

(b) Factorise $(x + y)^2 + 3(x + y)$

Let u = x + y. Then $u^2 + 3u = u(u + 3)$.

So $(x + y)^2 + 3(x + y) = (x + y)(x + y + 3)$.

1 mark
Factor out x + y

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(1)

(Total for Question 13 is 3 marks)

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